

Precalc Lesson 1.9 Homework

Name _____

1. The function $t = f(s)$ represents the number of school busses, t , needed to transport s students to school. Interpret the meaning of the statement $f^{-1}(33) = 650$.

2. Let $f(x) = -\frac{1}{3}x + 11$. Write an equation for $g(x)$, the inverse of f .

3. A table of selected values is given for a one-to-one function, $g(x)$. What is $g^{-1}(1)$?

x	-4	-2	0	1	5	8
$g(x)$	10	8	-3	-1	-4	1

4. Let $m = f(t)$ represent the total money earned while working at a company for t years. Explain what information is given by the function f^{-1} . Be sure to include both the input and the output in your explanation.
5. If $(-9, 4)$ is on the graph of f , which of the following ordered pairs is on the graph of f^{-1} ?
- A) $(9, -4)$
- B) $(-9, 4)$
- C) $(4, -9)$
- D) $(-4, 9)$

6. Consider the function $c = f(d) = d^3 + 5$.
- What is the independent variable of f^{-1} ?
 - What is the dependent variable of f^{-1} ?
 - Find f^{-1} .
7. A sink is filled with 54 quarts of water. The function $q = f(t) = 54 - 2.5t$ gives the volume of water in a sink, in quarts, t seconds after removing the stopper.
- Find f^{-1} .
 - What does the input of f^{-1} represent? What does the output of f^{-1} represent?
 - How long will it take for the sink to drain completely?
 - How long will it take for the sink to lose one-third of its water?
8. Let $g(x) = \sqrt{2x - 6}$.
- Find g^{-1} .
 - Use compositions to verify that g and g^{-1} are in fact inverses.